

Bhanu Pratap Singh

Software Engineer • AI/ML • Full-Stack Development | +91-7351707486 | bhanu2314578@gmail.com | Mumbai, MH | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Software Engineer with 1+ years of experience in AI/ML and full-stack development, specializing in building AI-powered automation systems and scalable web applications. Skilled in Python, React.js, Flask, NLP, and Machine Learning, with a proven track record of delivering solutions that drive real business impact.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, TypeScript, C, C++, SQL
- **Frontend:** React.js, Next.js, HTML5, CSS3, Tailwind CSS
- **Backend & APIs:** Flask, Node.js, Express.js, REST APIs, OAuth 2.0
- **AI / ML / NLP:** Supervised & Unsupervised ML, Deep Learning, NLP, BERT, TF-IDF, Word2Vec, TensorFlow (Keras), LLMs, Groq API, OpenAI API, Prompt Engineering, Reinforcement Learning (Q-Learning, SARSA, Policy Gradients)
- **Data & Analytics:** Pandas, NumPy, Matplotlib, Seaborn, Power BI, Tableau, Scikit-learn, Folium
- **Databases:** MongoDB, SQLite3, SQL **Other:** OCR, Web Scraping, Streamlit, Time Series (ARIMA, ADF Test, ACF, PACF)
- **DevOps & Tools:** Git, GitHub, CI/CD Pipelines, Docker (basic), VS Code

PROFESSIONAL EXPERIENCE

Associate Software Engineer (Bhatia Brothers FZE • Mumbai, MH)

Oct 2025 – Present

- Built an AI-powered Cost Estimation Agent using **NLP**, **OCR**, and **transformer-based** parsing to automate data extraction from emails, RFQs, and BOQs — achieved **85–90%** accuracy and reduced manual effort by **40%**.
- Developed predictive cost and profitability models using **regression** and **ensemble learning** to automate job costing sheets, improving pricing accuracy by **20%** and cutting turnaround from hours to minutes.
- Integrated Gmail ingestion with **OAuth 2.0** and optimized inference pipelines, reducing manual email handling by **60%** and processing latency by **30%**.

Full-Stack Developer Intern (Apna Samaj Tech Ltd. • Remote)

May 2023 – Jul 2023

- Developed scalable web applications using **Next.js**, **React.js**, and **TypeScript**, reducing code redundancy by **20%** and improving performance by **25%** via **lazy loading** and **code-splitting**.
- Implemented SEO optimization strategies, boosting search rankings by **15%** and reducing page load time by **30%**.
- Built and integrated **RESTful APIs** for frontend-backend sync, cutting API response time by **40ms**.

KEY PROJECTS

AI-Powered Learning Roadmap | [Github](#)

Jan 2025

- Built an AI-driven platform generating learning roadmaps by aggregating from platforms using **concurrent pipelines**.
- Integrated Groq LLM API with **Flask backend** and multi-source scraping; achieved **<2s response time** and **95%+** uptime.
- Developed responsive frontend with React + Tailwind with **caching** and **parallel fetching** for high performance.

Redundant Requirements Detection System (NLP Research) | [Github](#)

Dec 2024

- Built semantic similarity detection using **BERT embeddings**, **TF-IDF** and **Word2Vec** for SRS requirement deduplication.
- Achieved **95%** accuracy; published research demonstrating significant improvement over classical **NLP baselines**.
- Designed full pipeline: **tokenization**, **embedding generation**, and **similarity scoring**.

Agrotech Platform – Full-Stack AgriTech Solution | [Github](#)

Nov 2024

- Led full-stack development for land leasing, crop recommendation, and agro-product sales connecting farmers and buyers.
- Implemented district-specific crop prediction models with **scikit-learn**, **Flask APIs**, and **MongoDB**.
- Delivered **15%** higher crop yield and **10%** reduction in post-harvest losses for pilot users.

EDUCATION

B.Tech in Computer Science Ajay Kumar Garg Engineering College, Ghaziabad

Jul 2024 **CGPA: 8.07/10**

12th Standard (HSC) St. Fidelis, Aligarh, UP

May 2019 **87.4%**

10th Standard (SSC) Our Lady Of Fatima, Aligarh, UP

Jun 2017 **CGPA: 9.6/10**

CERTIFICATIONS

Python for Data Science, AI & Development	PW Skills	Sep 2023
Prompt Engineering & Programming with OpenAI	Columbia+	Sep 2023